



TEST EDITION

THE TEXTBOOK OF PHYSICS

For Grade 9



Sindh Textbook Board, Jamshoro

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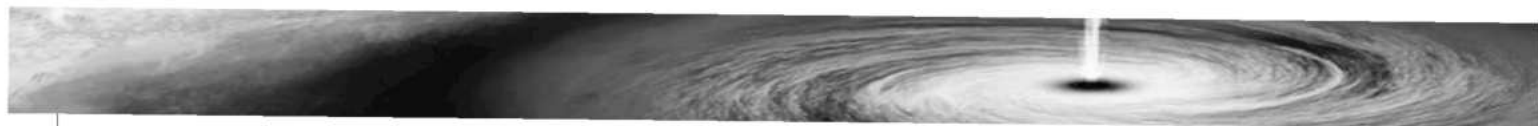
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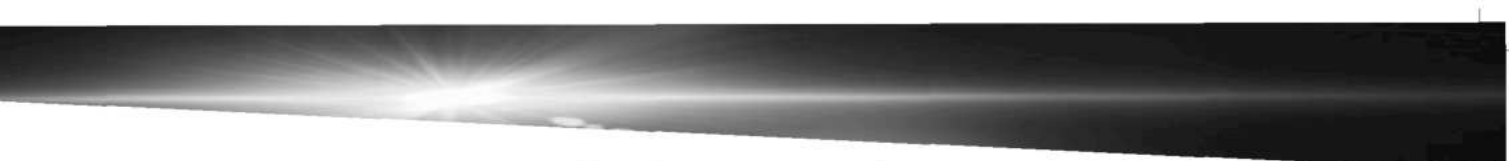
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PREFACE

The century we have stepped in, is the century of Physics. The modern disciplines Physics are strongly influencing not only all the branches of science but each and every aspect human life.

To keep the students abreast with the recent knowledge, it is a must that the curricula, at all the levels, be updated regularly by introducing the rapid and multidirectional development taking place in all the branches of Physics.

The recent book of Physics for class IX has been written in this preview and in accordance with the revised curriculum prepared by Ministry of Education, Govt of Pakistan, Islamabad reviewed by independent team of Bureau of Curriculum, Jamshoro Sindh. Keeping in view of the importance of Physics, the topics have been revised and re-written according to the need of the time.

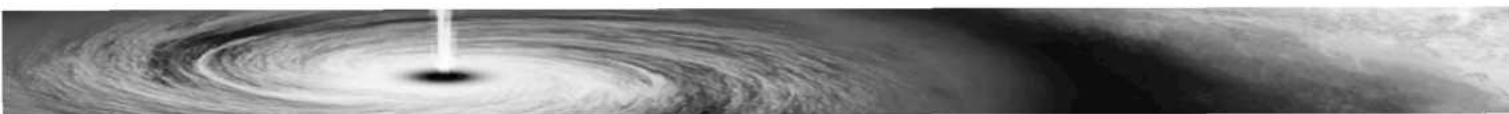
Since long Physics was teaching only in X class, the text book was consists of 18 units which was unable to complete in working hours. It has been decided now the Physics syllabus will be divided into portions, one should teach in 9th class and other will teach 10th class. So this book is consist of 9 units which have been thoroughly revised and re-written to meet the requirement of the curriculum.

Among the new editions are the introductory paragraphs, information boxes, summaries and a variety of extensive exercises which I think will not only develop the interest but also add a lot to the utility of the book.

The Sind Textbook Board has taken great pains and incurred expenditure in publishing this book inspite to its limitations. A textbook is indeed not the last word and there is always room for improvement. While the authors have tried their level best to make the most suitable presentation, both in terms of concept and treatment, there may still have some deficiencies and omissions. Learned teachers and worthy students are, therefore, requested to be kind enough to point out the short comings of the text or diagrams and to communicate their suggestions and objections for the improvement of the next edition of this book.

In the end, I am thankful to our learned authors, editors and specialist of Board for their relentless service rendered for the cause of education.

Chairman
Sindh Textbook Board



Unit - 1

Physical Quantities And Measurement

Nature is described as a pragmatic set of rules followed by all the things around us. It is something which is much greater than the imagination of humans. It is observable, it is surprising but it is somehow explainable, its architecture has been designed with very beautiful patterns, strict rules but with simplicity. A science which explores the nature is Physics.

Students Learning Outcomes (SLOs)

After learning this unit students should be able to:

- Describe the crucial role of Physics in Science, Technology and Society
- List with brief description of various branches of physics
- Choose a proper instrument (meter rule, Vernier calipers, screw gauge, physical balance stop watch, measuring cylinder) for the measurement of length, diameter, mass, time and volume in daily life activities.
- Interconvert the prefixes and their symbols to indicate multiple and sub-multiple for both base and derived units
- Write the answer in scientific notation in measurements and calculations
- Define term density with S.I unit
- Determine density of solids and liquids
- Describe the need of using significant figures for recording and stating results in laboratory.

$$E=mc^2$$



Quote

"No one undertakes research in physics with the intention of winning a prize. It is the joy of discovering something no one knew before."

Stephen Hawking



Do You Know!

Physics Derived from Ancient Greek 'physicos' meaning 'knowledge of nature'.

Why do we study physics? Which device will you choose to measure the length of a small cylinder? How will you determine the thickness of a piece of wire? How will you find the volume of small stone? why ice floats while a coin sinks in the water? After learning this unit you will be answer these and other similar questions.

1.1 INTRODUCTION TO PHYSICS

One of the most basic and ancient science is the Physics. The word science refers to the study of a fact by collecting information through observation, presenting it in a mathematical way, justifying the idea with experiment and finally making a conclusion about the fact. Thus physics can be defined as:

Physics is the branch of science which observes the nature represents it mathematically and conclude with the experiment.

It basically deals with the behavior and structure of matter and the energy that derives the matter. Physics is the branch of natural science that studies matter, its motion, its behavior through space and time and the related entities of energy and force. Physics is one of the most fundamental scientific disciplines, and its main goal is to understand how the universe behaves.

It is a matter of fact that Physics can be considered as the mother of all sciences. The beauty of physics lies in its Laws that govern this whole universe from an atom to large scale galaxies and in its experiments from home to large scale experiment labs. Physicist are categorized into two categories: those who observe the nature solve its mysteries with available



and missing information, present their theories with mathematical approach. They are known as theoretical physicist and other are more interested to test those theories with experiments are known as experimental physicists.

Since from the beginning of the universe, the structure of universe is very straight forward, the classification of physics was not that much easy but as the physicist explained the universe, they classified Physics into many branches. These branches show the spectrum and scope of Physics around us and help scientist to describe ideas in a well-organized way.

The main branches of Physics are as follows.

Mechanics

This branch of physics is mainly concerned with the laws of motion and gravitation.

Thermodynamics

Thermodynamics deals with heat and temperature and their relation to energy and work.

Electricity

Electricity is the study of properties of charges in rest and motion

Magnetism

Magnetism is the study of magnetic properties of materials

Atomic Physics

Atomic physics deals with the composition structure and properties of the atom



Fig. 1.1 Mechanics



Fig 1.2 Thermodynamics



Fig 1.3 Electricity



Fig 1.4 Magnetism

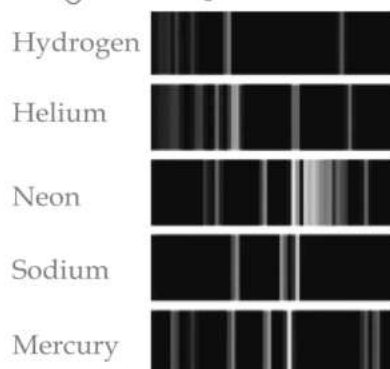


Fig. 1.5 Atomic Physics



Fig. 1.6 Optics



Fig. 1.7 Sound



Fig. 1.8 Nuclear Physics

Optics

Optics studies physical aspects of light and its properties with the help of optical instruments.

Sound

Sound is the study of production, properties and applications of sound waves.

Nuclear physics

Nuclear physics deals with the constituents, structure, behavior and interactions of atomic nuclei.

Particle physics

Particle Physics studies the elementary constituents of matter and radiation, and the interactions between them.

Astrophysics

The study of celestial objects with the help of laws of physics is known as **Astrophysics**.

Plasma physics

The study of ionized state of matter and its properties is known as **Plasma Physics**.

Geo physics

The study of internal structure of earth is known as **Geo physics**.

Importance of Physics in Science Technology and Society

Society's reliance on technology represents the importance of physics in daily life. Many aspects of modern society would not have been possible without



the important scientific discoveries made in the past. These discoveries became the foundation on which current technologies were developed.

Discoveries such as magnetism, electricity, conductors and others made modern conveniences, such as television, computers, smart phones, medical instruments, other business and home technologies possible. Moreover, modern means of transportation, such as aircraft and telecommunications, have drawn people across the world closer together all rely on concepts of physics.

1.2 MEASURING INSTRUMENTS

Physics is much concerned with matter and energy and the interaction between them which is explained with the help of describing the mathematical relations between various physical quantities. All **physical quantities** are important for describing the nature around us. A physical quantity is a physical property of a phenomenon, body, or substance that can be quantified by measurement.

A physical quantity can be expressed as the combination of a magnitude expressed by a number – usually a real number – and a unit. Physical quantities are classified into two categories:

- ◆ Fundamental quantities
- ◆ Derived physical quantities.

Physical quantities which cannot be explained by other physical quantities are called fundamental physical quantities.

There are seven fundamental physical quantities and are listed in table 1.1 along with their units.

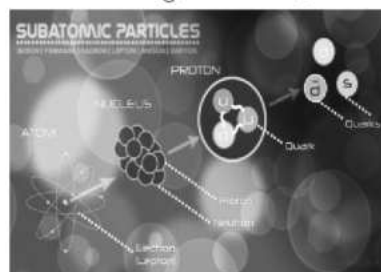


Fig. 1.9 Particle Physics



Fig. 1.10 Astro Physics

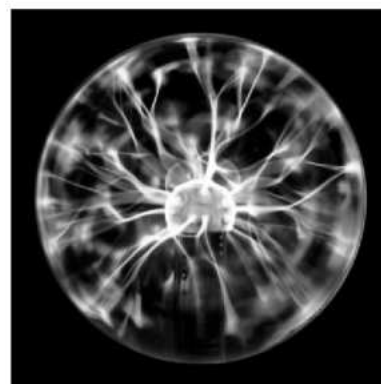


Fig. 1.11 Plasma Physics



Fig. 1.12 Geo Physics



Table 1.1 Fundamental quantities and their S.I units

Fundamental quantities	S.I Unit	Symbol of Unit
Length	meter	m
Mass	Kilogram	kg
Time	second	s
Electric current	Ampere	A
Temperature	Kelvin	K
Amount of substance	mole	mol
Luminous intensity	candela	cd



Do You Know!

Some Physical quantities are unit-less. Such as Elastic modulus, Plane angle and solid angle

Physical quantities which are explained on the basis of fundamental physical quantities are called derived physical quantities.

Table 1.2 derived quantities and their units

Derived Quantities	S.I Unit	Symbol of Unit
Volume	cubic meter	m^3
Velocity	meter per second	m^{-1}s
Force	Newton	N
Density	kilogram per cubic meter	kg/m^3
Acceleration	meter per second square	m/s^2



Do You Know!

The notion of physical dimension of a physical quantity was introduced by Joseph Fourier in 1822 by convention, physical quantities are organized in a dimensional system built upon base quantities, each of which is regarded as having its own dimension.

All physical quantities are either calculated mathematically or measured through an instrument. Scientist, Engineers, Doctors and others like blacksmith, carpenter, and goldsmith even the workers and ordinary human's measure those physical quantities with the help of instruments. For instance, your doctor uses a thermometer to tell your body temperature, a carpenter uses the inch tape to measure the length of woods required for furniture.



A puncture mender uses air gauges to check the air pressure in the tyre. Similarly, a chemical engineer uses hydrometer for describing the density of a liquid.

Measuring the physical quantity correctly with instrument is not an easy task for scientist and engineers. Scientist are seriously concerned with the accuracy of the instrument and its synchronization. Moreover, the instrument they design mostly for their own sake of research which readably goes on to commercial market. Many of the instruments we use today are inventions of pioneers of science. Usually, the basic physical quantities that we use in our daily life are measured with basic and simple instruments.

The Standard of Length

If there is any measurement that has proven to be the most useful to humanity, it is length. For examples units of length include the inch, foot, yard, mile, meter etc.

The length is defined as the minimum distance between two points lying on same plane.

The **meter** (m) is the SI unit of **length** and is defined as:

The length of the path traveled by light in vacuum during the time interval of $1/299\,792\,458$ of a second.

The basic measurement of length can be obtained with the help of a meter rod or an inch tape.

Meter Rule

A **meter rule** is a device which is used to measure length of different objects. A meter rule of length 1m is equal to 100 centimeters (cm). On meter rule each cm is divided further in to 10 divisions which



Do You Know!

Use of every instrument is restricted by smallest measurement that it can perform which is called least count.



Do You Know!

$1000\text{m} = 1\text{km}$
 $100\text{cm} = 1\text{m}$
 $1\text{cm} = 10\text{mm}$
 $1\text{inch} = 2.53\text{cm}$
 $12\text{ inch} = 1\text{ ft}$
 $1\text{ yard} = 3\text{ft}$



Fig 1.13 Meter Rule



Fig 1.14 Vernier Calipers



Fig 1.15 Digital vernier calipers

are called millimeters (mm). So, a meter rule can measure up to 1mm as smallest reading. It is made up of a long rigid piece of wood or steel(Fig 1.13).

The zero-end of the meter rule is first aligned with one end of the object and the reading is taken where the other end of the object meets the meter rule.

Vernier Caliper

The **Vernier Caliper** is a precision instrument that can be used to measure internal and external distance extremely accurate. It has both an imperial and metric scale. A Vernier caliper has main jaws that are used for measuring external diameter, as well as smaller jaws that are used for measuring the internal diameter of objects. Some models also have a depth gauge. The main scale is fixed in place, while the Vernier scale is the name for the sliding scale that opens and closes the jaws (Fig1.14).

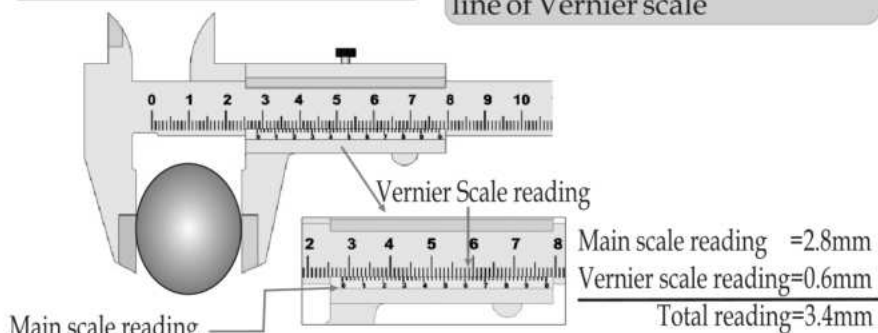
Reading a Vernier Caliper

Step 1

Place the object between the jaws of the Vernier caliper

Step 2

Note the main scale reading by counting lines before the zero line of Vernier scale



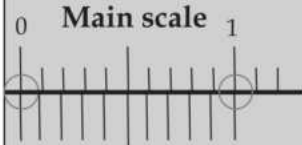
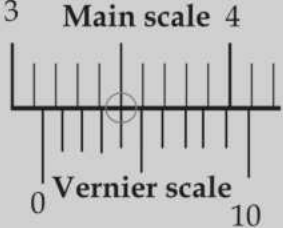
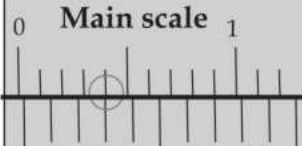
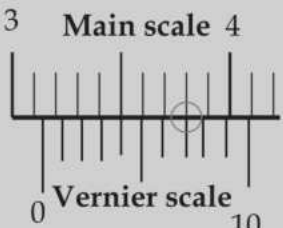
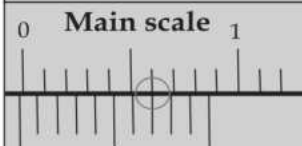
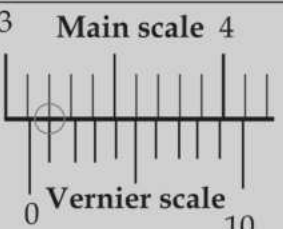
Step 3

Count the next line of Vernier scale after zero coinciding main scale

Step 4

Add the two reading for total



CHECKING FOR ZERO ERROR	OBSERVED READING	CORRECTED READING
 Two zero marks coincide No Zero error.	 Reading=3.14cm	3.14cm (No zero error No correction required)
 zero mark on vernier scale is slightly to the right Zero error is 0.03	 Reading=3.17cm	3.17cm- (+0.03)=3.14cm (The positive zero error is subtracted from reading)
 Zero mark on vernier scale is slightly to the left. zero error of -0.07	 Reading=3.11cm	3.11cm -(-0.07) =3.18cm (Negative zero error is added to the reading)

Micrometer Screw Gauge

Screw gauge is extensively used in engineering field for obtaining precision measurements. Micrometer screw gauge is used for measuring extremely small dimensions.

A screw gauge can even measure dimensions smaller than those measured by a Vernier Caliper. Micrometer Screw gauge works on the simple principle of converting small distances into larger ones by



Fig 1.16 Screw Gauge

measuring the rotation of the screw. This “screw” principle facilitates reading of smaller distances on a scale after amplifying them (Fig 1.16).

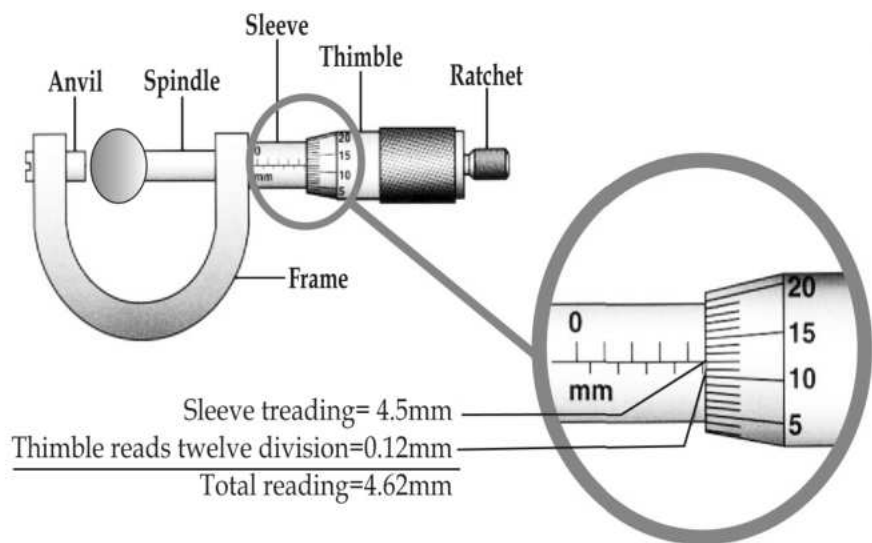
Reading A Micrometer Screw Gauge

Step 1

Turn the thimble until the anvil and the spindle gently grip the object. Then turn the ratchet until it starts to click.

Step 2

Take the main scale reading at the edge of the thimble.



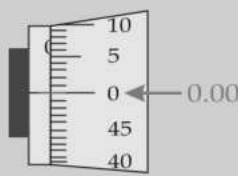
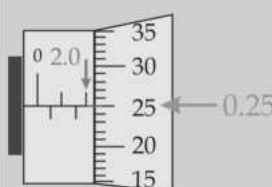
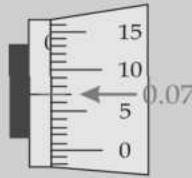
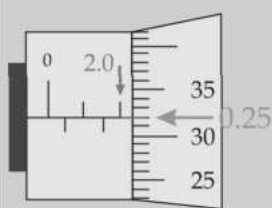
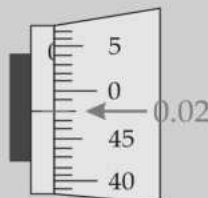
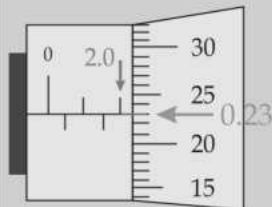
Step 3

Take the thimble scale reading opposite the datum line of the main scale. Multiply this reading with least count i.e., 0.01 mm

Step 4

Now add main scale reading to thimble reading. This will be the diameter of the object.



Checking For Zero Error	Observed reading	Corrected Reading
 Zero mark on thimble scale coincides with the datum line on the main scale and reading on the main scale is zero. No zero error	 Reading = $2.0 + 0.25$ = 2.25mm	2.25mm No zero error No Correction is required
 Zero on datum line can be seen. Positive Zero Error Reading = +0.07 mm (Count from Zero.)	 Reading = $2.0 + 0.32$ = 2.32mm	$2.32 - (+0.07)$ = 2.25mm
 Zero mark on datum line cannot be seen negative zero error Reading = -0.02mm (count down from 0)	 Reading = $2.0 + 0.23$ = 2.23mm	$2.23 - (-0.02)$ = 2.25mm



Do You Know!

The kilogram, originally defined as: The mass of one cubic decimeter of water at the temperature of maximum density. It was replaced after the International Metric Convention in 1875 by the International Prototype Kilogram.

The Standard Of Mass

The **kilogram** is the SI unit of **mass** and is equal to the mass of the international prototype of the kilogram, a platinum-iridium standard that is kept at the International Bureau of Weights and Measures (Fig1.17).



Fig 1.17 Kilo gram

**Do You Know!**

$$1000\text{g} = 1\text{kg}$$

$$1\text{g} = 1000\text{mg}$$

$$1\text{g} = 1000000\mu\text{g}$$

$$1\text{g} = 1000000000\text{ng}$$

$$1\text{g} = 0.002\text{lb}$$

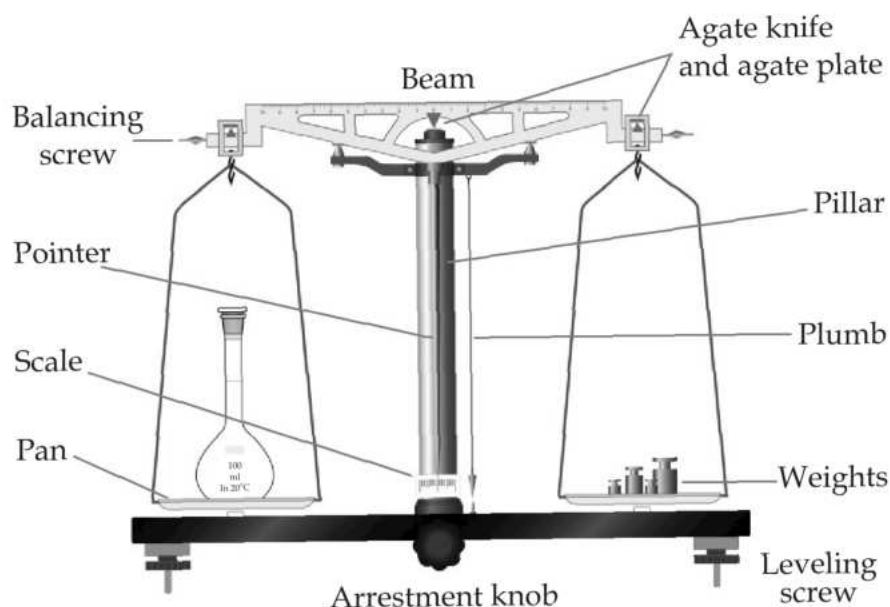
The kilogram is a cylinder of special metal about 39 millimeters wide by 39 millimeters tall that serves as the world's mass standard.

Each country that subscribed to the International Metric Convention was assigned one or more copies of the international standards; these are known as National Prototype Meter and Kilogram.

The Physical Balance

Fig 1.18 Physical Balance

The **Physical balance** is an instrument used for measurement of mass. It is mostly used in laboratory. It works on the principle of moments. It consists of a light and rigid beam of brass, a metallic pillar, a wooden base, two pans, a metallic pointer and an ivory scale (Fig 1.18). The plumb line indicates whether the balance is horizontal. In ideal condition the plumb line is aligned with the end of the knob fixed with the pillar. When the beam is horizontal the pointer remains on zero mark on the ivory scale. The whole box has leveling screws at the bottom to set it to horizontal. The device is enclosed in a glass box to avoid wind effects.





The Electronic Balance

The digital mass meter is an electronic instrument configured with integrated circuits and it works on the principal of balancing the forces.

The device is turned on and set to zero then object is placed on the plate. The reading on the screen gives the mass of object. The electronic balance (Fig 1.19) is available in different ranges of measurement such as micro gram, milligram and kilogram etc.



Fig 1.19 Electronic Balance

The Standard of Time

Before 1960, the standard of time was defined in terms of the mean solar day for the year 1900. The rotation of the Earth is now known to vary slightly with time, this motion is not a good one to use for defining a time standard.

In 1967, the **second** was redefined to take advantage of the high precision attainable in a device known as an atomic clock(Fig 1.20), which uses the characteristic frequency of the cesium-133 atom as the “reference clock”.

The second is now defined as 9 192 631 770 times the period of vibration of radiation from the cesium atom.



Fig 1.20 Atomic Clock

Stop Watch

A stopwatch is used to measure the time interval between two events. There are two types of stopwatch : Mechanical stopwatch and Digital stopwatch.

Mechanical/ Analogue Stopwatch

A **mechanical stop** watch can measure a time interval up to 0.1 second (Fig1.21). It has a knob that is



Fig. 1.21 Stop Watch



Fig. 1.22 Digital stop watch

used to wind the spring that powers the watch. It can also be used as a start stop and reset button. The watch starts when the knob is pressed once. When pressed second time, the watch stops. While the third press brings the needle back to zero.

Digital Stopwatch

A digital stop watch can measure a time interval up to 0.01 second (fig 1.22). It starts to indicate the time lapsed as the start/stop button is pressed. As soon as start/stop button is pressed again, it stops and indicates the time interval recorded by it between start and stop of an event. A reset button restores its initial zero setting. Now a days almost the mobile phones have a stopwatch function.

Human Reaction Time

As analogue or digital or watch is operated by human manually i.e., they have to be started or stopped by hand. This causes a random error in measurement of time i.e called human reaction time. For most people human reaction time is about 0.3- 0.5 s. Therefore for more accurate measurement of time intervals light gates (Fig1.23) can be used.

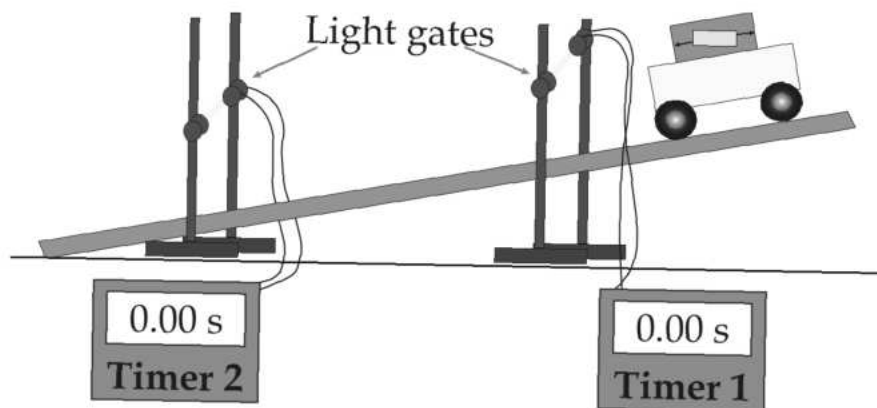


Fig 1.23 Light gates



SELF ASSESSMENT QUESTIONS:

Q1: What instrument will you choose to measure height of your friend?

Q2: Can you describe how many seconds are there in a year?

Q3: Which instrument will you choose to measure your mass?



Do You Know!

1 hour = 60 min

1 hour = 3600 sec

1min=60sec

1sec=1000ms

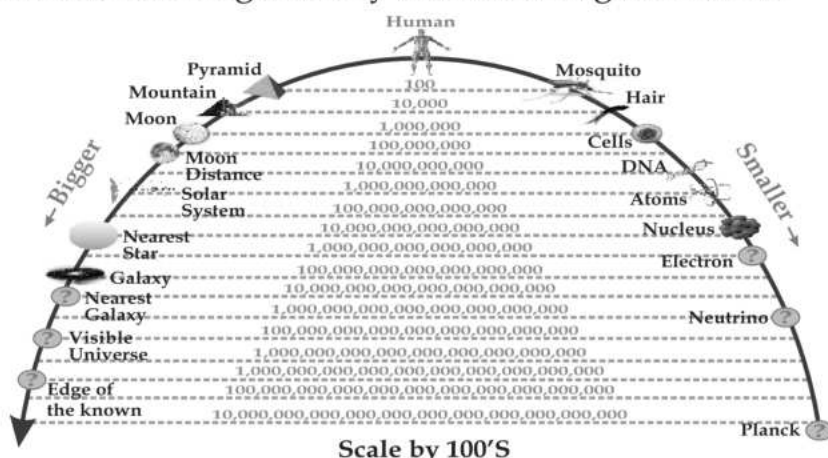
1sec=1000000 μ s

1.3 PREFIXES

The Physical quantities are described by the scientist in terms of magnitudes and units. Units play a vital role in expressing a quantity either base or derived. Prefixes are useful for expressing units of physical quantities that are either very big or very small.

A **unit prefix** is a specifier. It indicates multiples or fractions of the units.

Units of various sizes are commonly formed by the use of such prefixes. The prefixes of the **metric system**, such as kilo and milli, represent multiplication by powers of ten. Historically, many prefixes have been used or proposed by various sources, but only a narrow set has been recognized by standards organizations.



Scale by 100'S
Table 1.3 SI pre fixes



SI Prefixes

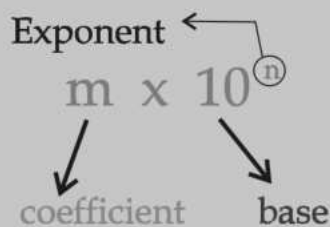
Prefix	Symbol	Meaning	Multiplier (Numerical)	Multiplier (Exponential)
<i>Greater than 1</i>				
tera	T	trillion	1 000 000 000 000	10^{12}
giga	G	billion	1 000 000 000	10^9
mega	M	million	1 000 000	10^6
kilo	k	thousand	1 000	10^3
hecto	h	hundred	100	10^2
deka	da	ten	10	10^1
<i>Less than 1</i>				
Unit	1			
*deci	d	tenth	0.1	10^{-1}
*centi	c	hundredth	0.01	10^{-2}
*milli	m	thousandth	0.001	10^{-3}
*micro	μ	millionth	0.000 001	10^{-6}
*nano	n	billionth	0.000 000 001	10^{-9}
pico	p	trillionth	0.000 000 000 001	10^{-12}
femto	f	quadrillionth	0.000 000 000 000 001	10^{-15}
atto	a	quintillionth	0.000 000 000 000 000 001	10^{-18}

SELF ASSESSMENT QUESTION:

Q4: Can you tell if the size of a nucleus is up to 10^{-15} m.
What prefix shall we use to describe its size?

1.4 SCIENTIFIC NOTATION

Scientific Notation



Scientific notation or the standard form is a simple method of writing very large numbers or very small numbers. In this method numbers are written as powers of ten. Thus calculation of very large or very small numbers becomes easy.

Numbers in Scientific Notation are made up of three parts: The coefficient, the base and the exponent.

- ◆ The coefficient must be equal to or (Not zero) greater than one
- ◆ The base must be 10
- ◆ The exponent can be negative or positive.



However, a gas has neither a fixed shape nor a fixed volume- it will expand to fill its container.

Often we find the large weight woods floating on the surface of water. However, an iron needle sinks into the water. We say iron is “heavier” than wood. This cannot really be true rather we should say like iron is “denser” than wood. Physicist are concerned with a physical quantity, a property of matter which may help to define the nature of matter in terms of its mass and space.



Measuring the Volume

For density to be measured or calculated we first need to find the volume of substances. Most of solid geometrical shapes have formulae for their volume which is obtained through different parameters such as radius, height, depth, width, base and length, but for irregular objects, liquids and gases this approach is unusual. The volume of liquids can be measured with the help of Cylinders, and Beakers.

Measuring Cylinder

Measuring cylinder is a glass or plastic cylinder with a scale-graduated in cubic centimeters or milliliters (ml)(fig1.24). It is used to find the volume of liquids. When a liquid is poured, it rises to a certain height in the cylinder. The level of liquid in the cylinder is noted and volume of the liquid is obtained.

In order to read the volume correctly we should keep the eye in level with the bottom of the meniscus of the liquid surface as you learned in previous grade.

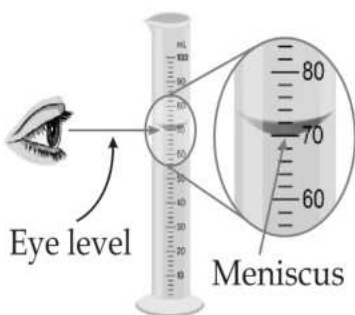


Fig. 1.24
Measuring Cylinder



1. Volume of Liquid

A volume of about a liter or so can be measured using a measuring cylinder. When the liquid is poured into the cylinder the level on scale gives the volume. Most measuring cylinders have scales marked in milliliters (ml) or cubic centimeters (cm^3). It should be noted that while recording the value from cylinder the eyes should maintain the level with the value. Angular observation may result a false reading of the volume.

2. Regular solid

If an object has a regular shape its volume can be calculated

For instance:

Volume of a rectangular block = length $\times$ width $\times$ height

Volume of a cylinder = $\pi \times \text{radius}^2 \times \text{height}$

3. Irregular solid

For an irregular solid its volume is calculated by lowering the object in a partially filled measuring cylinder (fig 1.25). The rise in the level on the volume scale gives the volume of that object. Thus the volume of irregular solid is calculated by subtracting the original volume of liquid from the raised volume of liquid.

The total volume is found. The volume of the solid is measured in a separate experiment and then subtracted from the total volume.



Quick Lab

Take a measuring cylinder of 1 liter capacity at full place it in a beaker.

Fill cylinder full with water.

Pour a stone of irregular shape in it gradually.

As you pour the stone in the cylinder, the water from cylinder drops into the beaker.

Drop the stone in cylinder completely

Calculate the volume of water ejected out of cylinder.

Volume of water ejected is the volume of the stone.

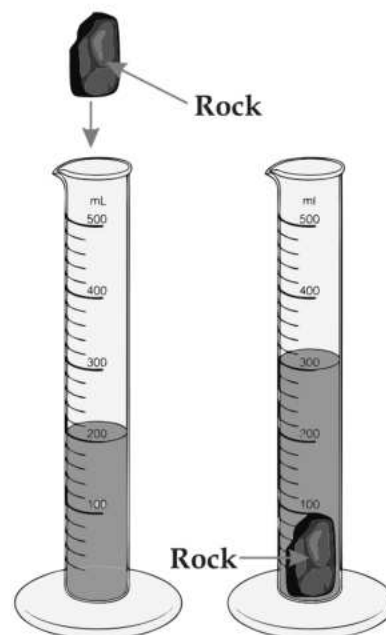


Fig 1.25. Volume Irregular shaped Solid

**Do You Know!**

During the Cold War between Russia and America. There was a race of Astrophysics. America was facing the period of racism. A Black lady mathematician named Katherine solved the problem of putting the first orbital satellite.

Recommended!

Watch movie "Hidden Figures"
Observe the importance of Reliable Numbers.

Density

The term **density** of a substance is defined as mass of substance (m) per unit volume (V). It is denoted by Greek letter **ρ (rho)**.

$$\rho = \frac{m}{V}$$

Density is characteristic property of any pure substance. Objects made of a particular pure substance such as pure Gold can have any size or mass but its density will be same for each.

In accordance with the above equation mass of a substance can be expressed as

$$m = \rho V$$

The S.I unit for density is kg/m^3 kgm^{-3} . Sometimes dens of substances is given in gm/cm^3 . The density of Aluminum is 2.70 gm/cm^3 which is equal to 2700 Kg/m^3 .

**Do You Know!**

In Jordan there is sea known as 'Dead Sea'

The humans in that sea while swimming does not sink!

This is because the water of sea is much more salty than normal, which raises the density of water.

Measuring the Density

It is to be noted that there are two ways of finding the density of a substance either mathematically or experimentally by taking density of water at 4°C as a reference which is sometimes known as relative density or '**Specific gravity**'. It has no unit, it is a number whose value is the same as that of the density in g/cm^3 .

$$\text{relative density} = \frac{\text{density of substance}}{\text{density of water}}$$



Worked Example 4

What is the mass a solid iron wrecking ball of radius 18cm. if the density of iron is 7.8 gm/cm^3 ?

Solution:

Step 1: write known physical quantities with units and point out the quantity to be found.

Density of iron ball $\rho = 7.8 \text{ gm/cm}^3 = 7.8 \times 1000 \text{ kg/m}^3$

Radius of iron ball is $r = 18\text{cm} = 18 \times 10^{-2} \text{ m} = 0.18\text{m}$

Volume of the iron ball is $V = (4/3) \times \pi \times r^3 = (1.33) \times 3.14 \times (0.18\text{m})^3 = 0.024\text{m}^3$

Step 2: write down the formula and rearrange if necessary

$$m = \rho \times V$$

Step 3: put the values in formula and calculate

Since mass of iron ball is $m = \rho \times V = (7.8 \times 10^3) \times (0.024)$

$$m = 187.2 \text{ kg}$$

SELF ASSESSMENT QUESTIONS:

Q5: How can you identify which gas is denser among the gases?

Q6: Can you tell how hot air balloon works?

1.6 SIGNIFICANT FIGURES

Engineers and scientist around the world work with numbers either representing a large or small magnitude of a physical quantity. The engineers are however interested in the accuracy of a value as they mostly work on estimation but scientist especially physicist are more concerned in the accuracy of these numbers. For instance, an engineer records the speed of wind and explains it on an average. On the other hand, for the physicist, the speed of earth on its course, the



speed of light in vacuum the mass or charge on an electron is just not a matter of numbers but accurate numbers.

The numbers of reliably known digits in a value are known as significant figures.

Table 1.4 Rules for determining significant figures

Rule	Example
1. All non-zeroes are significant	2.25 (3 significant figures)
2. Leading zeroes are NOT significant	0.00000034 (2 significant figures)
3. Trailing zeroes are significant ONLY if an explicit decimal point is present	200 (1 significant figure) 200. (3 significant figures) 2.00 (3 significant figures)
4. Trapped zeroes are significant	0.00509 (3 significant figures) 2045 (4 significant figures)

Worked example 5

How many significant figures are there in the area of a cylinder whose diameter is 5 cm

Solution:

Step 1: write known physical quantities and point out the unknown quantity

Diameter of the cylinder is $d = 5\text{cm} = 5 \times 10^{-2}\text{m} = 0.05\text{m}$

Radius of cylinder is $r = d/2 = 2.5 \times 10^{-2}\text{m} = 0.025\text{m}$

Step 2: write down formula and rearrange if necessary

The area of the cylinder is $A = \pi \times r^2 = 3.14 \times (0.025\text{m})^2 = 0.0019\text{m}^2$

Step 3: put value in formula and calculate

Thus area of cylinder can be written as $A = 1.9\text{mm}^2$

Thus, there are two significant numbers in the value 1 and 9.



SELF ASSESSMENT QUESTIONS:

Q7: Determine the number of significant figures in 00.6022009



SUMMARY

- ◆ Physics is the branch of science which deals with studies of matter its composition, properties, and interaction with energy.
- ◆ The branches of Physics are classified on the basis of different areas of study with different approaches.
- ◆ There are two types of physicist, theoretical and experimental physicist.
- ◆ Physics define mathematical relation between physical quantities. A physical quantity has magnitude and unit.
- ◆ Physical quantity are mainly classified into two categorize
 - (i) Base or fundamental quantities
 - (ii) Derived physical quantities.
- ◆ Base quantities are length, mass, time, temperature, current, luminous intensity, and amount of substance.
- ◆ The standard of length is meter can be measured by measuring tape, or meter rule.
- ◆ The standard of mass is kilogram can be measured by physical balance.
- ◆ The standard of time is second can be measured by stop watch.
- ◆ The measured or calculated values either macroscopic or microscopic can be expressed in Scientific Notations.
- ◆ The volume of liquid is calculated or measured with help of measuring cylinder

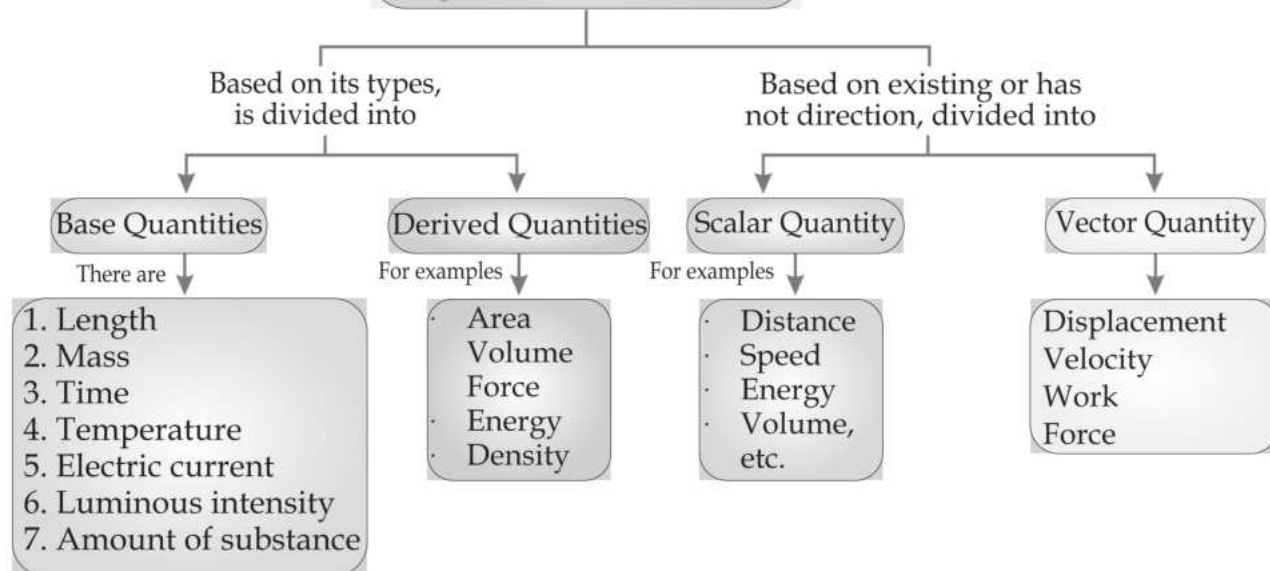


- ◆ The volume of irregular objects can be calculated through measuring cylinder with displacement of water.
- ◆ The density of a pure substance is its characteristic property it is the ratio of mass per unit volume.
- ◆ The density of objects can be calculated with the help of water as a reference known as specific gravity also known as relative density.
- ◆ Prefixes can be used to represent large or smaller values of a physical quantity.
- ◆ The most accurate or reliable numbers of a value are known as significant figures.

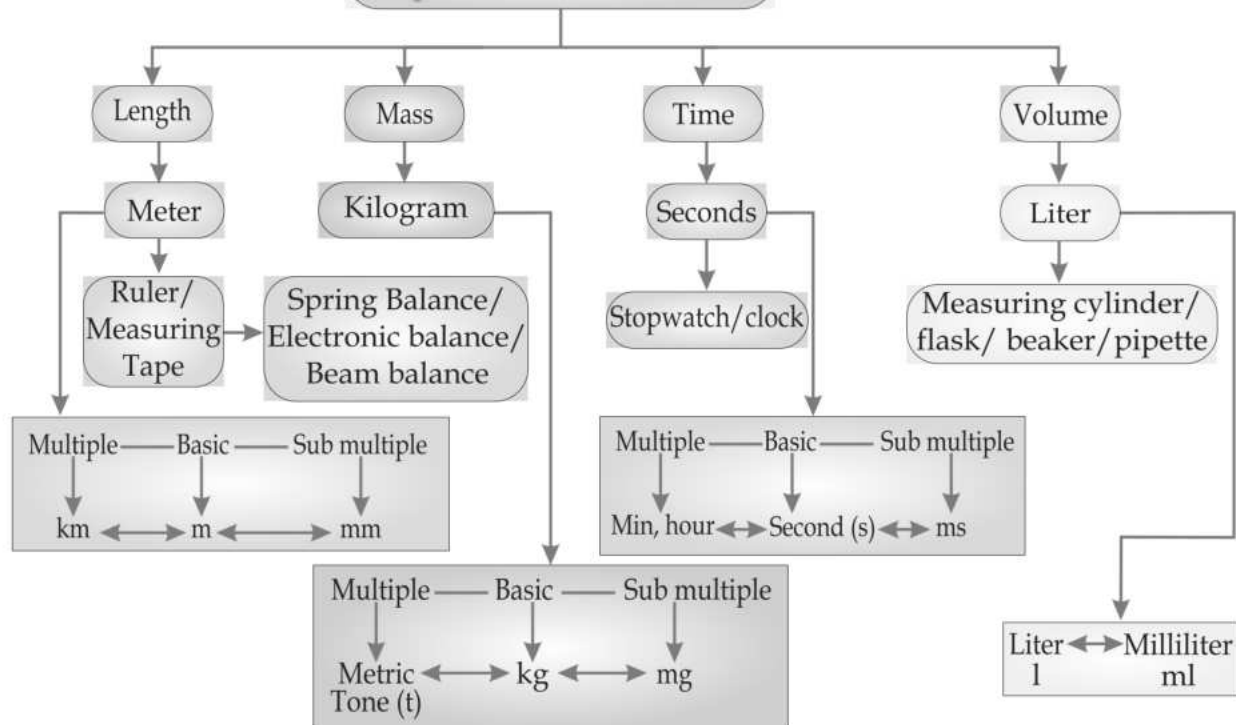


CONCEPT MAP

Physics Quantities



Physics Quantities





End of Unit Questions

Section (A) Multiple Choice Questions (MCQs)

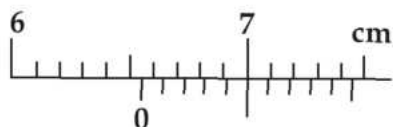


Fig 1.26

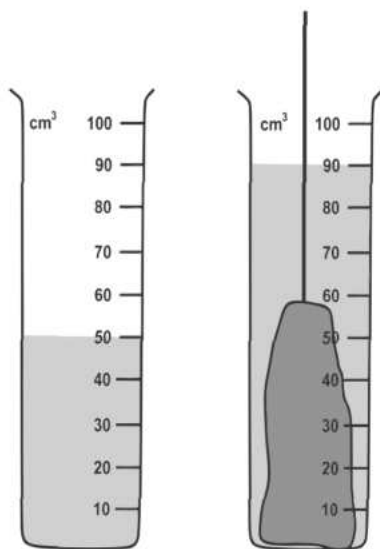


Fig 1.27

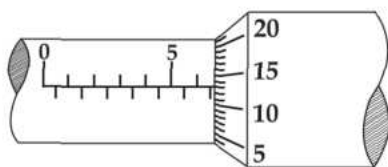


Fig 1.28

- The Figure 1.26 shows part of a Vernier scale, what is the reading on the Vernier scale
 - 6.50 cm
 - 6.55 cm
 - 7.00 cm
 - 7.45 cm
- Ten identical steel balls each of mass 27g, are immersed in a measuring cylinder having 20cm³ of water. The reading of water level rises to 50cm³. What is the density of the steel?
 - 0.90 gm/cm³
 - 8.1 gm/cm³
 - 9.0 gm/cm³
 - 13.5 gm/cm³
- An object of mass 100g is immersed in water as shown in the figure 1.27, what is the density of the material from which object is made?
 - 0.4gcm³
 - 0.9gcm³
 - 1.1 gcm³
 - 2.5gcm³
- What is the reading of this micrometer in figure 1.28
 - 5.43mm
 - 6.63mm
 - 7.30mm
 - 8.13mm
- A chips wrapper is 4.5 cm long and 5.9 cm wide. Its area upto significant figures will be
 - 30 cm²
 - 28 cm²
 - 26.55 cm²
 - 32 cm²



6. A worldwide system of measurements in which the units of base quantities were introduced is called
- prefixes
 - international system of units
 - hexadecimal system
 - none of above
7. All accurately known digits and first doubtful digit in an expression are known as
- non-significant figures
 - significant figures
 - estimated figures
 - crossed figures
8. If zero line of Vernier scale coincides with zero of main scale, then zero error is
- | | |
|-------------|---------|
| a) positive | b) zero |
| c) negative | d) one |
9. zero error of the instrument is
- | | |
|---------------------|---------------------|
| a) systematic error | b) human error |
| c) random error | d) classified error |
10. Length, mass, electric current, time, intensity of light and amount of substance are examples of
- | | |
|--------------------|------------------------|
| a) base quantities | b) derived quantities |
| c) prefixes | d) quartile quantities |



Section (B) Structured Questions

1.

Column A Action	Column B Branch
Cooking Bar B.Q	Thermodynamics
Turning the Bulb on	
Riding a bicycle	
Looking for Giant Galaxies	
Producing a loud sound	
Describing an atom	
Obtaining energy from Earth	

2.

Physical Quantity	S.I Unit	Type
Ampere		
	m^3	
	Sec	Base
Temperature		Base
	N	
Density	Kg per m^3	
Acceleration		

3. Convert the following values.

- 230 cm = _____m
- 250 g = _____kg
- 0.5 s = _____ms
- 0.8 m = _____mm
- 350ms = _____s
- 1.2Kg = _____g



4. An engineer measures the width of an aluminum sheet using Vernier caliper as shown in fig 1.29

a) What is the measurement of the width of aluminum sheet

b) Which gives more precise measurement Vernier caliper, Screw Gauge or meter rule?

5. A pendulum swings as shown in figure 1.30 from X to Y and back to X again

i) What would be the most accurate way of measuring time for one oscillation? with the help of a Stop Watch.

a) Record time for 10 oscillations and multiply by 10

b) Record time for 10 oscillation and divide by 10

c) Record time for one oscillation

d) Record time from X to Y and double it

ii) Suggest an instrument for measuring time period more accurately.

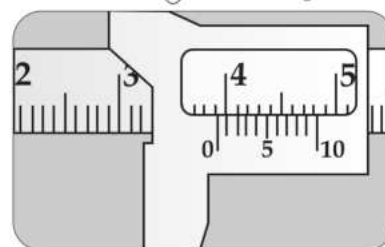


Fig 1.29

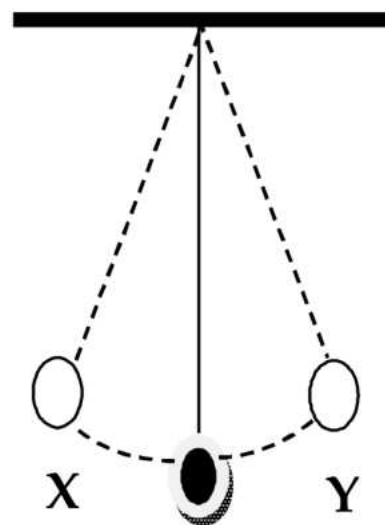


Fig. 1.30

Prefixes

6. write the correct prefix of notion

a) $75000\text{m} = 750$ _____

b) $2/1000\text{ sec} = 1$ _____

c) $1/1000000\text{ g} = 1$ _____

d) $1000000000\text{ m} = 1$ _____

Scientific Notation

7. Write values in standard and scientific notation

a) The radius of 1st orbit of Hydrogen atom is $r = 0.53 \text{ \AA} =$ _____

b) 1 light year is $2628000000000\text{m} =$ _____

c) Vacuum pressure $2.7 \times 10^{-4}\text{ torr} =$ _____



Density and Volume

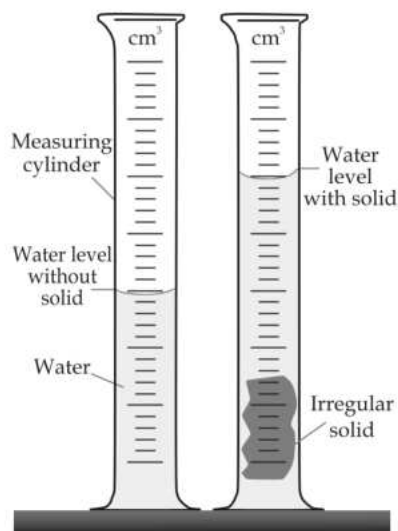


Fig 1.31

8. A wooden piece is made in different shapes take length (l) = radius (r) = 2m Calculate its volume as a:
- | | |
|-------------|------------|
| a) Sphere | b) Cube |
| c) Cylinder | d) Pyramid |
| e) Cylinder | |
9. Find the density of wood as sphere and cube if the mass of wood is 1kg. Is there any change in density due to shape?
10. A measuring cylinder (fig 1.31) is filled with 500cc water. A stone of mass 20g is immersed in to the cylinder such that ,water level rises up to 800cc. Which statement is correct?
- | |
|--|
| a) The difference between the readings gives the density of stone. |
| b) The difference between the readings gives volume of the stone |
| c) The final reading gives the density of stone |
| d) The final reading gives the volume of stone |

Significant Figures

11. Write significant numbers in the following values.
- | |
|---|
| a) 980 has _____ Significant numbers. |
| b) 91.60 has _____ Significant numbers. |
| c) 10010.100 has _____ Significant numbers. |
| d) 0.0086 has _____ Significant numbers. |